

IDA / 3MTT Data Analysis

Capstone Project Compendium

Six Real-World Projects Integrating
Excel • Power BI • SQL • Python

Each project includes:

Full Dataset | Objectives | Task Breakdown | Timeline | Rubric

2026

Introduction & Instructions

This compendium contains six capstone projects for students who have completed the IDA/3MTT data analysis programme covering Microsoft Excel, Power BI, SQL, and Python. Each project is based on a real-world business scenario and comes with a purpose-built dataset. **You are expected to pick any 3 out of the 6 to work on.**

How Projects Are Structured

Every project follows the same format:

- Overview — Background context and business problem
- Dataset Description — Column definitions and data dictionary
- Learning Objectives — What skills each project develops
- Tasks — Step-by-step breakdown across all four tools
- Deliverables — Exactly what to submit
- Grading Rubric — Clear criteria with scores

Tool Usage Requirements

Every project **MUST** use all four tools:

Tool	Minimum Requirement	Purpose
Python	EDA notebook with at least 6 visualizations	Cleaning, exploration, automation
SQL	At least 10 queries including JOINS, GROUP BY, window functions	Data storage & querying
Excel	Pivot tables, charts, formulas, and a summary sheet	Quick analysis & reporting
Power BI	Interactive dashboard with at least 4 visuals and 2 slicers	Stakeholder presentation

General Submission Requirements

1. Create a folder for your project clearly named folder: YOUR_FELLOW_ID_project if you are a 3mtt fellow or LASTNAME_FIRSTNAME_project if you are a regular IDA student.
2. In that folder, create 3 different sub-folders (folders inside the one created in step 1), each of the sub-folders should be name base on the project you are picking, each subfolders will contain the submission for that project. Each project requires the following:
3. Python: Submit a .ipynb Jupyter notebook with all outputs visible
4. SQL: Submit a .sql file with all queries and comments
5. Excel: Submit the .xlsx file with all sheets clearly labeled

6. Power BI: Submit a .pbix file and a PDF export of the dashboard
7. Include a 1-2 page written summary (PDF or Word) interpreting your findings
8. All visuals must have titles, axis labels, and appropriate formatting
9. Zip your project folder and submit via <https://forms.gle/c2cpiCh8pYWexrT3A>.
10. Note that the submission link will remain valid June 05, 2026, 11:59:00 PM GMT+1 and you can only make 1 submission. Please confirm your files before submitting.
11. If you are not sure of anything, feel free to ask questions on the WhatsApp group chat

Grading Scale

Total Score	Grade	Description
90 – 100	A (Distinction)	Outstanding work; exceeds all requirements
75 – 89	B (Merit)	Strong work; meets all requirements with quality
60 – 74	C (Pass)	Satisfactory; meets most requirements
50 – 59	D (Borderline)	Meets minimum requirements with gaps
Below 50	F (Fail)	Does not meet the minimum requirements

PROJECT 1

Retail Sales Performance Analysis

Domain: E-Commerce / Retail | Dataset: retail_sales.csv (80,000 rows)

Business Background

ShopEase Nigeria is a mid-sized retail company operating across eight major cities in Nigeria. The company sells products across eight categories through both physical stores and an online platform. Management has noticed that while overall revenue has grown, some regions and product categories are consistently underperforming. They have hired you as a data analyst to investigate trends, understand customer behaviour, and produce a comprehensive performance dashboard.

Dataset Description

File: retail_sales.csv | Rows: 80,000 | Period: 2021 – 2023

Column	Type	Description
order_id	Text	Unique identifier for each order
customer_id	Text	Unique customer identifier
customer_name	Text	Full name of the customer
customer_email	Text	Customer email address
product_name	Text	Name of the product purchased
category	Text	Product category (8 categories)
region	Text	City/region of purchase
quantity	Integer	Number of units ordered
unit_price	Float	Price per unit in Naira
discount_rate	Float	Discount applied (0 to 0.20)
net_revenue	Float	Revenue after discount
payment_method	Text	Payment channel used
order_date	Date	Date of the order (YYYY-MM-DD)
delivery_days	Integer	Number of days to deliver
returned	Text	Whether the order was returned (Yes/No)
salesperson_id	Text	ID of the salesperson

Learning Objectives

- Perform end-to-end data cleaning and exploratory analysis in Python
- Write SQL queries to answer business KPI questions
- Build pivot-table summaries and trend charts in Excel
- Design an interactive sales dashboard in Power BI
- Identify actionable insights from sales patterns

Tasks

Part A — Python (Data Cleaning & EDA)

12. Load the dataset and display basic info (shape, dtypes, nulls, duplicates)
13. Convert order_date to datetime and extract year, month, quarter, day of week
14. Calculate a new column: gross_revenue = quantity × unit_price; compare to net_revenue
15. Plot monthly revenue trends as a line chart across all three years
16. Create a bar chart showing top 10 products by total revenue
17. Visualize return rates by product category using a horizontal bar chart
18. Analyse payment method distribution using a pie/donut chart
19. Compute and visualize correlation between delivery_days and return rate
20. Build a heatmap showing average revenue by region and category
21. Write a brief markdown summary (in the notebook) of your 5 key findings

Part B — SQL (Business Queries)

22. Create a table called retail_sales and import the dataset
23. Total revenue by region — ordered highest to lowest
24. Top 5 customers by total spending
25. Monthly revenue trend: year, month, total revenue, total orders
26. Return rate by category (% of orders returned)
27. Average delivery days by payment method
28. Revenue contribution by category (category, revenue, % of total)
29. Identify customers who have placed more than 5 orders
30. Month-over-month revenue growth using a window function (LAG)
31. Products with above-average revenue per unit
32. Rank salespersons by total revenue using RANK()

Part C — Excel (Summary & Pivot Analysis)

33. Import the CSV into Excel and format as a table
34. Create a Pivot Table: Revenue by Region and Category (cross-tab)
35. Create a Pivot Table: Monthly orders and revenue trend with a line chart
36. Use SUMIF/COUNTIF to calculate total revenue and orders for each payment method

37. Calculate return rate per category using a formula column
38. Create a bar chart: Top 10 products by net revenue
39. Highlight the top 20% of customers by revenue using conditional formatting
40. Build an executive summary sheet with key KPIs using formula-linked cells

Part D — Power BI (Dashboard)

41. Import the CSV and set up data types correctly in Power Query
42. Create a date table and establish relationships
43. KPI Cards: Total Revenue, Total Orders, Average Order Value, Return Rate
44. Bar chart: Revenue by Region
45. Donut chart: Revenue by Category
46. Line chart: Monthly Revenue Trend
47. Matrix: Region × Category revenue breakdown
48. Add slicers for Year, Region, Category, and Payment Method
49. Add a page-level filter for returned orders
50. Format the dashboard with consistent colours, titles, and logo area

Timeline

Week	Focus	Tasks
Week 1	Python	Load data, clean, feature engineering, EDA plots 1–5
Week 2	Python + SQL	Complete EDA (plots 6–10), set up SQL database, write queries 1–5
Week 3	SQL + Excel	Complete SQL queries 6–11, Excel pivot tables and charts
Week 4	Power BI + Report	Build Power BI dashboard, write summary report, review and submit

Grading Rubric

Criteria	Excellent (4)	Good (3)	Fair (2)	Poor (1)	Score
Python — Code Quality & Cleaning	Clean, well-commented code; all null/duplicate issues resolved	Minor issues; mostly clean	Some errors; partial cleaning	Code has major bugs or no cleaning	/4
Python — Visualizations	6+ charts, all properly labelled with insights	5 charts with labels	3–4 basic charts	Fewer than 3 or unlabelled	/4

SQL — Query Accuracy	All 11 queries correct and optimized	8–10 correct	5–7 correct	Fewer than 5 correct	/4
SQL — Advanced Techniques	Uses JOINS, window functions, CTEs correctly	Uses window functions	Basic GROUP BY only	Only simple SELECTs	/4
Excel — Pivot Tables	All pivots correct with formatted charts	Most pivots correct	Some pivots; minimal formatting	Missing pivots	/4
Power BI — Dashboard Design	Professional layout; 4+ visuals; slicers functional	Good design; 3 visuals	Basic dashboard; 1–2 visuals	Incomplete or non-functional	/4
Insights & Interpretation	Deep, actionable insights with supporting evidence	3–4 good observations	1–2 surface observations	No meaningful insights	/4
Submission Quality	All files submitted; naming convention followed	Minor missing item	2 items missing	Major parts missing	/4
TOTAL					/32

PROJECT 2

Hospital Patient Flow & Resource Analysis

Domain: Healthcare | Dataset: hospital_patient_flow.csv (60,000 rows)

Business Background

Lagos General Hospital is a busy tertiary care facility that serves patients from across Lagos and neighbouring states. Hospital management has been concerned about long patient waiting times, high readmission rates, and revenue collection gaps. The Chief Medical Director has commissioned a data analysis project to understand patient flow, identify high-demand periods, and generate insights to guide resource allocation, staffing, and billing improvements.

Dataset Description

File: hospital_patient_flow.csv | Rows: 60,000 | Period: 2021 – 2023

Column	Type	Description
patient_id	Text	Unique patient identifier
age	Integer	Patient age in years
gender	Text	Male or Female
state_of_origin	Text	Patient's state of origin
department	Text	Hospital department (8 departments)
diagnosis	Text	Primary diagnosis (16 diagnoses)
admission_date	Date	Date of admission
admission_time	Time	Time of admission (HH:MM)
discharge_date	Date	Date of discharge
length_of_stay_days	Integer	Days spent in hospital
outcome	Text	Discharge outcome (5 categories)
insurance_type	Text	Type of insurance coverage
total_bill_ngn	Float	Total bill amount in Naira
amount_paid_ngn	Float	Amount actually paid
attending_doctor_id	Text	Doctor ID
ward	Text	Ward number
readmitted	Text	Whether patient was readmitted (Yes/No)

Learning Objectives

- Analyse healthcare operational data to identify inefficiencies
- Calculate clinical KPIs including occupancy rate, readmission rate, and collection rate
- Use SQL to query complex multi-condition patient records
- Build an operational dashboard for hospital management

Tasks

Part A — Python

51. Load the dataset; check for missing values and duplicates
52. Parse admission and discharge dates; compute a derived column: $\text{bill_gap} = \text{total_bill} - \text{amount_paid}$
53. Plot patient admission volume by month and year (line chart)
54. Bar chart: Top 5 departments by number of admissions
55. Stacked bar chart: Outcomes by Department
56. Box plot: Length of Stay by Department
57. Heatmap: Admissions by hour of day vs day of week (to find peak periods)
58. Pie chart: Insurance type distribution
59. Bar chart: Average bill vs average payment by insurance type
60. Scatter plot: Age vs length of stay coloured by outcome
61. Summary: 5 operational insights derived from the analysis

Part B — SQL

62. Create and populate a `hospital_patients` table
63. Total admissions and average length of stay per department
64. Monthly admissions trend with year-over-year comparison
65. Top 10 diagnoses by frequency
66. Readmission rate per department (% of patients readmitted)
67. Revenue gap: total billed vs total collected per insurance type
68. Peak admission hours: admissions grouped by hour of day
69. Patients with length of stay greater than 14 days — list details
70. Mortality rate (outcome = 'Deceased') by department and year
71. Rank departments by average bill amount using `RANK()`
72. CTE: Identify the top 5 wards by patient load each year

Part C — Excel

73. Import data and format as a table
74. Pivot: Admissions by month and department
75. Pivot: Average length of stay by diagnosis
76. Calculate collection rate per insurance type ($\text{amount_paid} / \text{total_bill}$)
77. Conditional formatting: Highlight patients with $\text{bill_gap} > \text{N}100,000$

- 78. Create a column chart: Top diagnoses by admission count
- 79. Create a line chart: Monthly admission trends
- 80. Executive KPI sheet: Total admissions, avg LOS, readmission rate, collection rate

Part D — Power BI

- 81. Import dataset and create a proper date hierarchy
- 82. KPI Cards: Total Admissions, Avg Length of Stay, Readmission Rate, Collection Rate
- 83. Bar chart: Admissions by Department
- 84. Line chart: Monthly Admissions Trend
- 85. Donut chart: Insurance Type Split
- 86. Table visual: Revenue billed vs collected by insurance type
- 87. Slicers: Year, Department, Outcome, Insurance Type
- 88. Drill-through page: Patient detail view for a selected department

Timeline

Week	Focus	Tasks
Week 1	Python	Data loading, cleaning, date parsing, visualizations 1–6
Week 2	Python + SQL	Remaining charts, derived columns, SQL queries 1–6
Week 3	SQL + Excel	SQL queries 7–11, all Excel pivots and KPI sheet
Week 4	Power BI + Report	Dashboard build, drill-through page, written report, submit

Grading Rubric

Criteria	Excellent (4)	Good (3)	Fair (2)	Poor (1)	Score
Python — Cleaning & Feature Engineering	bill_gap computed; all dates parsed; nulls handled	Most features correct	Partial feature work	No derived columns	/4
Python — Clinical Visualizations	10 charts with clinical context and annotations	7–9 charts with labels	4–6 charts	Fewer than 4	/4
SQL — Query Accuracy	All 11 queries correct	8–10 correct	5–7 correct	Below 5	/4
SQL — Advanced (CTE, Window Fns)	Correct CTE and RANK usage	One advanced query	Only basic queries	No advanced SQL	/4

Excel — Pivots & Formatting	All pivots correct; conditional formatting applied	Most pivots; some formatting	Some pivots; no formatting	Incomplete	/4
Power BI — Drill-through & Slicers	Drill-through functional; 4+ slicers	Drill-through or slicers	Basic dashboard	Non-functional	/4
Healthcare Insights	Specific operational recommendations with evidence	3 good insights	1–2 general observations	No insights	/4
Report Quality	Clear, concise, professional 1–2 page summary	Good summary	Brief and incomplete	No report	/4
TOTAL					/32

Ride-Hailing Operations Analysis

Domain: Transport / Gig Economy | Dataset: ridehailing_trips.csv (100,000 rows)

Business Background

SwiftRide is a ride-hailing startup operating in Lagos. Having completed its second year of operations, the company wants to review its performance data to optimise driver allocation, understand demand patterns, improve customer satisfaction, and reduce cancellation rates. The operations team has provided you with a full trip log covering 100,000 rides over three years and expects a comprehensive analysis with actionable recommendations.

Dataset Description

File: ridehailing_trips.csv | Rows: 100,000 | Period: 2021 – 2023

Column	Type	Description
trip_id	Text	Unique trip identifier
driver_id	Text	Driver identifier
rider_id	Text	Rider identifier
pickup_zone	Text	Pickup location (15 Lagos zones)
dropoff_zone	Text	Drop-off location
trip_date	Date	Date of the trip
trip_hour	Integer	Hour of day (0–23)
trip_day_of_week	Text	Day of the week
car_type	Text	Vehicle category (Economy, Comfort, XL, Premium, Moto)
distance_km	Float	Trip distance in kilometres
duration_minutes	Integer	Trip duration in minutes
fare_ngn	Float	Fare charged in Naira
surge_multiplier	Float	Surge pricing applied (1.0 = no surge)
payment_method	Text	Cash, Card, or Wallet
driver_rating	Float	Customer rating of driver (3.0–5.0)
trip_status	Text	Completed, Cancelled, or No-Show
cancellation_reason	Text	Reason if cancelled

Learning Objectives

- Analyse time-series trip data to identify demand patterns
- Calculate driver performance KPIs and segment performance
- Build geospatial and temporal visualizations in Python
- Use SQL window functions to rank drivers and analyse trends
- Design an operational dashboard for real-time monitoring

Tasks

Part A — Python

89. Load data, inspect structure, fix dtypes, check nulls
90. Feature engineering: $\text{revenue_per_km} = \text{fare_ngn} / \text{distance_km}$; time_of_day bucket (Morning/Afternoon/Evening/Night)
91. Line chart: Daily trip volume over time
92. Bar chart: Trips by hour of day (showing demand peaks)
93. Stacked bar: Trip status distribution by car type
94. Heatmap: Avg fare by pickup zone and time of day bucket
95. Bar chart: Top 10 pickup zones by completed trip count
96. Scatter plot: Distance vs Fare, coloured by surge multiplier
97. Bar chart: Average driver rating by car type
98. Line chart: Monthly cancellation rate trend
99. Summary: 5 operational recommendations

Part B — SQL

100. Create and populate a trips table
101. Total trips, total revenue, avg fare per car type
102. Cancellation rate per zone (pickup_zone)
103. Busiest hours: trips grouped by hour, sorted descending
104. Driver performance table: trips completed, avg rating, total revenue per driver
105. Top 10 drivers by total revenue this year
106. Zone-pair analysis: most popular pickup-dropoff combinations
107. Surge pricing impact: avg fare with surge vs without
108. Weekly revenue trend using DATE_TRUNC or STRFTIME
109. Riders with 10+ completed trips (loyal riders)
110. Rolling 7-day average trip count using window functions

Part C — Excel

111. Import CSV and create a formatted table
112. Pivot: Total revenue and trips by zone
113. Pivot: Average fare by car type and day of week
114. Formula: cancellation rate = COUNTIF(Cancelled) / total trips

- 115. Line chart: Monthly revenue trend
- 116. Bar chart: Revenue by pickup zone
- 117. Conditional formatting: Highlight drivers with avg rating below 4.0
- 118. KPI sheet: Total Revenue, Completed Trips, Cancellation Rate, Avg Rating

Part D — Power BI

- 119. Import data and set up time intelligence with a date table
- 120. KPI Cards: Total Revenue, Total Trips, Cancellation Rate, Avg Driver Rating
- 121. Bar chart: Trips by Zone (sorted)
- 122. Area chart: Daily trips trend
- 123. Donut chart: Trip status breakdown
- 124. Table: Top 20 drivers by revenue
- 125. Slicers: Car Type, Zone, Year, Month
- 126. Tooltip page showing zone-level stats on hover

Timeline

Week	Focus	Tasks
Week 1	Python	Load, clean, feature engineering, time-of-day bucketing, charts 1–5
Week 2	Python + SQL	Charts 6–11, SQL setup, queries 1–6
Week 3	SQL + Excel	SQL queries 7–11, all Excel tasks
Week 4	Power BI + Report	Dashboard, tooltip page, written summary, submit

Grading Rubric

Criteria	Excellent (4)	Good (3)	Fair (2)	Poor (1)	Score
Python — Feature Engineering	Both derived columns correct; time bucket accurate	One column correct	Partially correct	No feature engineering	/4
Python — Demand Analysis Plots	10 charts capturing time/zone/surge insights	7–9 charts	4–6 charts	Fewer than 4	/4
SQL — Trip & Driver Queries	All 11 queries return correct results	8–10 correct	5–7 correct	Below 5	/4
SQL — Window Functions	Rolling avg and ranking queries correct	One window function	Attempted but incorrect	No window functions	/4

Excel — Pivots & Formulas	All pivots and formulas correct; formatted	Most tasks done	Some tasks done	Minimal work	/4
Power BI — Tooltip & Slicers	Tooltip functional; 4 slicers; clean layout	Tooltip or slicers	Basic dashboard	Non-functional	/4
Operational Insights	5 specific, evidence-based recommendations	3–4 good insights	1–2 general points	No insights	/4
Presentation Quality	Professional formatting; all visuals titled	Good quality	Acceptable	Poor quality	/4
TOTAL					/32

PROJECT 4

Student Performance & Learning Outcomes Tracker

Domain: Education | Dataset: student_performance.csv (50,000 rows)

Business Background

The Lagos State Ministry of Education wants to understand academic performance trends across 100 secondary schools in the state. They are particularly concerned about high failure rates in certain subjects, performance gaps between public and private schools, and the impact of attendance on exam outcomes. You have been engaged to analyse three years of student assessment data, identify at-risk populations, and provide evidence-based recommendations for intervention.

Dataset Description

File: student_performance.csv | Rows: 50,000 | Period: 2021 – 2024 (3 academic years)

Column	Type	Description
student_id	Text	Unique student identifier
school_id	Text	Unique school identifier (100 schools)
lga	Text	Local Government Area of school
gender	Text	Male or Female
age	Integer	Student age
course	Text	Subject name (10 subjects)
term	Text	Term 1, 2, or 3
academic_year	Text	Academic year
attendance_pct	Float	Attendance percentage (0–100)
ca_score	Float	Continuous assessment score
exam_score	Float	Examination score
total_score	Float	Combined total score (0–100)
grade	Text	Letter grade: A, B, C, D, or F
passed	Text	Yes if total_score >= 45
study_hours_per_week	Float	Self-reported weekly study hours
parent_education	Text	Highest parental education level

school_type	Text	Public or Private
internet_access	Text	Yes or No

Learning Objectives

- Perform cohort analysis to identify at-risk student groups
- Use SQL to query complex educational data across multiple dimensions
- Build pivot-table summaries for school and subject performance
- Create a monitoring dashboard for education administrators

Tasks

Part A — Python

127. Load, inspect, and clean the dataset
128. Calculate pass rate per subject and visualize as a horizontal bar chart
129. Box plot: Score distribution by school type (Public vs Private)
130. Scatter plot: Attendance % vs Total Score with a regression line
131. Bar chart: Average score by LGA
132. Grouped bar chart: Pass rate by gender across subjects
133. Heatmap: Average score by subject and academic year
134. Bar chart: Score comparison by parental education level
135. Line chart: Failure rate trend across the three academic years
136. Correlation matrix: Attendance, study hours, CA score, exam score
137. Flag at-risk students: attendance below 60% AND total score below 45

Part B — SQL

138. Create and populate a student_performance table
139. Pass rate by subject — ordered from lowest to highest
140. Average score by school_type and gender (cross-tab using GROUP BY)
141. Schools with pass rate below 50% for Maths
142. Top 10 schools by overall pass rate
143. Subjects where female students outperform male students
144. Attendance impact: avg total score for attendance buckets (0–40, 41–60, 61–80, 81–100)
145. Students who passed all subjects in a given academic year
146. LGA performance ranking using RANK()
147. Improvement trend: avg score per subject comparing first and last academic year
148. CTE: Identify bottom 10% of students by average score

Part C — Excel

149. Import and format data as a table
150. Pivot: Pass rate by Subject and School Type
151. Pivot: Average score by LGA and Gender
152. AVERAGEIF formula: Avg score for students with attendance $\geq 80\%$
153. COUNTIF: Number of students with grade F per subject
154. Conditional formatting: Highlight all F grades in red
155. Bar chart: Pass rate by subject
156. Summary KPI sheet: Total students, overall pass rate, avg attendance, top-performing subject

Part D — Power BI

157. Import dataset; ensure grade and school_type are categorical
158. KPI Cards: Total Students, Overall Pass Rate, Avg Score, Avg Attendance
159. Clustered bar: Pass rate by Subject
160. Bar chart: Average score by LGA
161. Matrix: School performance by Term and Subject
162. Scatter plot: Attendance vs Score
163. Slicers: Academic Year, School Type, LGA, Gender, Subject
164. Bookmarks: Toggle between a 'Subject View' and 'School View'

Timeline

Week	Focus	Tasks
Week 1	Python	Data cleaning, at-risk flagging, charts 1–5
Week 2	Python + SQL	Charts 6–11, SQL table setup, queries 1–5
Week 3	SQL + Excel	SQL queries 6–11, all Excel tasks
Week 4	Power BI + Report	Dashboard, bookmarks, written report, submit

Grading Rubric

Criteria	Excellent (4)	Good (3)	Fair (2)	Poor (1)	Score
Python — At-Risk Identification	Correct flagging logic; explained in notebook	Mostly correct	Partial logic	No at-risk analysis	/4
Python — Educational Visualizations	10 labelled charts with educational insight	7–9 charts	4–6 charts	Fewer than 4	/4

SQL — Multi-Dimension Queries	All 11 queries correct and efficient	8–10 correct	5–7 correct	Below 5	/4
SQL — CTE & Ranking	Correct CTE for bottom 10%; LGA ranking correct	One correct	Attempted incorrectly	Not attempted	/4
Excel — Conditional Formatting	All grades highlighted; formulas correct	Most tasks done	Some tasks	Minimal	/4
Power BI — Bookmarks	Both bookmark views functional; slicers work	Bookmarks or slicers	Basic dashboard	Non-functional	/4
Policy Recommendations	3+ specific, evidence-backed recommendations	2 good insights	1 general suggestion	No recommendations	/4
Data Storytelling	Clear narrative connecting findings to policy	Good narrative	Disconnected points	No narrative	/4
TOTAL					/32

Personal Finance & Budget Analysis Tool

Domain: Consumer Finance | Dataset: personal_finance.csv (75,000 rows)

Business Background

FinTrack is a Nigerian fintech startup building a personal finance app. Before launching, the product team wants to analyse three years of synthetic transaction data from 5,000 test users to understand spending patterns, savings behaviour, and budgetary pressures. The findings will inform in-app budget recommendations, alert features, and marketing segmentation. You have been hired as the data analyst to produce a complete spending behaviour report and interactive dashboard.

Dataset Description

File: personal_finance.csv | Rows: 75,000 | Period: 2021 – 2023

Column	Type	Description
transaction_id	Text	Unique transaction identifier
user_id	Text	Unique user identifier (5,000 users)
transaction_date	Date	Date of transaction
month	Text	Month name
year	Integer	Year of transaction
transaction_type	Text	Income or Expense
category	Text	Spending or income category
amount_ngn	Float	Transaction amount in Naira
description	Text	Short transaction description
payment_method	Text	Payment method used
bank	Text	Bank or fintech platform
savings_goal_met	Text	Yes / No / N/A
budget_category	Text	Budget classification
notes	Text	Optional notes (Planned, Urgent, etc.)

Learning Objectives

- Aggregate and analyse financial transaction data across users and time

- Build a multi-user SQL financial database with meaningful queries
- Create a personal budget model in Excel
- Design a consumer-facing finance dashboard in Power BI

Tasks

Part A — Python

1. Load and inspect data; check for anomalies in amounts
2. Separate income and expense records; compute net savings per user per month
3. Bar chart: Total spending by category
4. Line chart: Monthly income vs expense trend (aggregated across all users)
5. Pie chart: Spending category share
6. Box plot: Monthly expense distribution by category
7. Bar chart: Average monthly savings by user income bracket
8. Bar chart: Top banks by transaction volume
9. Heatmap: Spending by category and month
10. Flag over-spenders: Users where expenses > income in more than 3 months
11. Summary: 5 insights for the product team

Part B — SQL

12. Create and populate a transactions table
13. Total income vs total expense per user (net position)
14. Average monthly spending by category
15. Users with a savings rate above 20% (savings / income)
16. Top spending category per month — use window function
17. Bank popularity: transaction count and volume by bank
18. Monthly net savings trend (all users combined)
19. Users who have never saved (no category = 'Savings' in their records)
20. Cumulative spending per user using SUM() OVER (ORDER BY date)
21. Month with highest average expense per user
22. CTE: Identify users in the top 10% of spenders

Part C — Excel

23. Import data and format as a table
24. Pivot: Total income and expenses per month
25. Pivot: Spending breakdown by category for each year
26. Build a budget model: Input monthly income → auto-calculate recommended category budgets using the 50/30/20 rule
27. Formula: Savings rate = (income - expenses) / income
28. Conditional formatting: Months where expenses exceed income highlighted in red
29. Bar chart: Category spending vs recommended budget

30. KPI sheet: Total income, total expenses, net savings, savings rate

Part D — Power BI

- 31. Import data; separate income vs expense using a calculated column
- 32. KPI Cards: Total Income, Total Expenses, Net Savings, Savings Rate
- 33. Donut chart: Spending by Category
- 34. Line chart: Monthly Income vs Expense
- 35. Bar chart: Top 10 spending categories
- 36. Table: User-level income, expenses, savings rate
- 37. Slicers: Year, Month, Transaction Type, Category, Bank
- 38. DAX measure: Savings Rate = (SUM(Income) - SUM(Expense)) / SUM(Income)

Timeline

Week	Focus	Tasks
Week 1	Python	Load, clean, income/expense split, charts 1–5
Week 2	Python + SQL	Charts 6–11, over-spender flag, SQL setup and queries 1–5
Week 3	SQL + Excel	SQL queries 6–11, budget model in Excel, KPI sheet
Week 4	Power BI + Report	Dashboard, DAX measure, summary, submit

Grading Rubric

Criteria	Excellent (4)	Good (3)	Fair (2)	Poor (1)	Score
Python — Income/Expense Separation	Correct split; net savings calculated per user per month	Mostly correct	Partial	Not done	/4
Python — Finance Visualizations	10 labelled charts with financial insight	7–9 charts	4–6 charts	Fewer than 4	/4
SQL — Financial Queries	All 11 correct; savings rate formula accurate	8–10 correct	5–7 correct	Below 5	/4
SQL — Cumulative / Window Fns	Cumulative SUM correct; top 10% CTE correct	One correct	Attempted	Not attempted	/4
Excel — 50/30/20 Budget Model	Functional model that auto-calculates on income change	Model works partially	Static formula only	No model	/4

Power BI — DAX Measure	Savings Rate measure correct; reflects slicer context	Correct but ignores slicers	Incorrect measure	No DAX	/4
Product Insights	5 actionable product recommendations	3–4 insights	1–2 general	None	/4
Dashboard Usability	Consumer-friendly; clear labels; intuitive layout	Good design	Basic	Hard to read	/4
TOTAL					/32

PROJECT 6

Agricultural Supply Chain Analysis

Domain: Agriculture / Agribusiness | Dataset: agric_supply_chain.csv (70,000 rows)

Business Background

AgroLink Nigeria is an agribusiness company that sources crops from farmers across 12 Nigerian states and distributes them to major markets nationwide. The company has been experiencing growing post-harvest losses, inconsistent pricing, and high logistics costs. The management team wants a comprehensive analysis of their supply chain data to identify where value is being lost, which crops and states deliver the best margins, and how storage, transport, and cooperative membership affect outcomes.

Dataset Description

File: agric_supply_chain.csv | Rows: 70,000 | Period: 2021 – 2023

Column	Type	Description
record_id	Text	Unique record identifier
farmer_id	Text	Unique farmer identifier (10,000 farmers)
crop	Text	Crop type (12 crops)
state	Text	State of origin (12 states)
lga	Text	Local Government Area
harvest_date	Date	Date of harvest
season	Text	Dry Season, Wet Season, or Off-Season
quantity_harvested_kg	Float	Total quantity harvested in kg
post_harvest_loss_pct	Float	Percentage lost post-harvest
quantity_sold_kg	Float	Quantity that reached market
farm_gate_price_per_kg	Float	Price at the farm gate (NGN/kg)
market_price_per_kg	Float	Market selling price (NGN/kg)
gross_revenue_ngn	Float	Total market revenue
transport_mode	Text	Mode of transport used
transport_cost_ngn	Float	Cost of transportation
destination_market	Text	Target market name
days_to_market	Integer	Days from harvest to market

storage_type	Text	Storage method used
fertilizer_used	Text	Yes or No
irrigation_used	Text	Yes or No
cooperative_member	Text	Yes or No (farmer cooperative)

Learning Objectives

- Analyse agricultural supply chain data across multiple dimensions
- Quantify post-harvest losses and their revenue impact
- Build SQL queries for margin and logistics analysis
- Design a supply chain performance dashboard

Tasks

Part A — Python

39. Load and inspect dataset; handle any data quality issues
40. Feature engineering: $\text{price_margin} = \text{market_price} - \text{farm_gate_price}$; $\text{net_profit} = \text{gross_revenue} - \text{transport_cost}$
41. Bar chart: Average post-harvest loss % by crop
42. Bar chart: Total gross revenue by state
43. Line chart: Average market price per kg by month and year
44. Grouped bar: Avg post-harvest loss — cooperative members vs non-members
45. Bar chart: Transport mode vs average days to market
46. Box plot: Price margin by crop
47. Heatmap: Avg net profit by state and crop
48. Bar chart: Revenue by storage type
49. Summary: 5 supply chain improvement recommendations

Part B — SQL

50. Create and populate an agric_supply_chain table
51. Total revenue and avg post-harvest loss by crop (sorted by loss DESC)
52. State performance: total revenue, avg margin, avg loss per state
53. Effect of cooperative membership on avg price and loss rate
54. Top 5 most profitable crop-state combinations
55. Transport cost as % of gross revenue by transport mode
56. Monthly price trend: avg farm gate price vs market price per crop
57. Farmers with post-harvest loss > 20% — count and avg revenue impact
58. Storage type impact: avg loss % and avg revenue by storage type
59. Market destination performance: revenue and volume per market
60. Rank crops by net profit margin using RANK() window function

Part C — Excel

- 61. Import and format data as a table
- 62. Pivot: Total revenue by crop and state
- 63. Pivot: Average post-harvest loss by crop and season
- 64. Formula: revenue lost to spoilage = quantity_harvested × loss% × farm_gate_price
- 65. Conditional formatting: Crops with loss > 15% highlighted in orange
- 66. Bar chart: Net profit by crop
- 67. Line chart: Monthly average market price for top 5 crops
- 68. KPI sheet: Total revenue, total transport costs, avg loss %, total farmers

Part D — Power BI

- 69. Import dataset; create a date hierarchy
- 70. KPI Cards: Total Revenue, Avg Post-Harvest Loss %, Total Transport Cost, Net Profit
- 71. Bar chart: Revenue by State
- 72. Bar chart: Post-harvest loss by Crop
- 73. Scatter plot: Days to Market vs Post-harvest Loss %
- 74. Matrix: Crop performance by Season
- 75. Slicers: Crop, State, Season, Storage Type, Cooperative Membership
- 76. Tooltip page: Farmer-level breakdown on hover

Timeline

Week	Focus	Tasks
Week 1	Python	Data cleaning, feature engineering, charts 1–5
Week 2	Python + SQL	Charts 6–11, loss quantification, SQL queries 1–5
Week 3	SQL + Excel	SQL queries 6–11, Excel model and loss formula
Week 4	Power BI + Report	Dashboard, tooltip page, recommendations, submit

Grading Rubric

Criteria	Excellent (4)	Good (3)	Fair (2)	Poor (1)	Score
Python — Feature Engineering	price_margin and net_profit correct; losses quantified	One column correct	Partial	Not done	/4

Python — Supply Chain Visualizations	10 labelled charts capturing losses, margins, and geography	7–9 charts	4–6 charts	Fewer than 4	/4
SQL — Margin & Loss Queries	All 11 queries correct; margin calculations accurate	8–10 correct	5–7 correct	Below 5	/4
SQL — RANK & Window Fns	RANK() on crops correct; price trend query accurate	One correct	Attempted	Not attempted	/4
Excel — Loss Revenue Formula	Formula correct; conditional formatting applied	Formula correct only	Static values	Not done	/4
Power BI — Scatter & Tooltip	Scatter plot formatted; tooltip functional	One of them	Basic dashboard	Neither works	/4
Business Recommendations	5 specific supply chain actions with evidence	3–4 recommendations	1–2 general	None	/4
Cross-Tool Consistency	Findings consistent across all 4 tools	Minor inconsistency	Some discrepancies	Major discrepancies	/4
TOTAL					/32

Quick Reference: Datasets at a Glance

#	Project	File	Rows	Size	Key Tools Focus
1	Retail Sales Performance	retail_sales.csv	80,000	~11 MB	Python EDA, SQL window fns, Power BI dashboard
2	Hospital Patient Flow	hospital_patient_flow.csv	60,000	~7.6 MB	Python clinical plots, SQL CTEs, PBI drill-through
3	Ride-Hailing Operations	ridehailing_trips.csv	100,000	~11.5 MB	Python time series, SQL rolling avg, PBI tooltips
4	Student Performance	student_performance.csv	50,000	~5.6 MB	Python regression, SQL ranking, PBI bookmarks
5	Personal Finance	personal_finance.csv	75,000	~9.2 MB	Python segmentation, SQL cumulative sums, DAX
6	Agricultural Supply Chain	agric_supply_chain.csv	70,000	~11 MB	Python heatmaps, SQL margin analysis, PBI scatter

Recommended Datasets by Skill Level

Level	Recommended Projects	Why
Beginner-Friendly	Projects 1 & 5	Clean data; clear business KPIs; straightforward SQL
Intermediate	Projects 3 & 4	Requires time-series logic; window functions; bookmarks
Advanced	Projects 2 & 6	Multiple derived columns; CTEs; drill-throughs; scatter

Tools Installation Checklist

- Python 3.x with: pandas, numpy, matplotlib, seaborn, jupyter
- SQL: MySQL, PostgreSQL, or SQLite (any is acceptable)
- Microsoft Excel 2016 or later (or Microsoft 365)
- Microsoft Power BI Desktop (free download from Microsoft)

Popsicool is saying Good luck to us all. Remember: the quality of your insights matters more than the volume of your code.